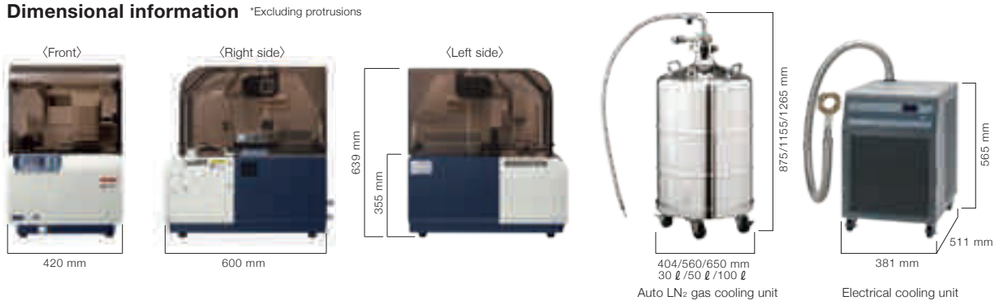
Primary Specifications

	DSC600	DSC200
Measurement principle	Heat-flux type	
Temperature range	-150°C to 725°C	
DSC dynamic range	±100 mW	±200 mW
RMS noise / Sensitivity	0.05 µW/0.1 µw	0.1 µW/0.2 µw
Calorimetric precision	±0.05% *Calculated from the standard deviation (1σ)/mean from Indium 10 consecutive measurements	
Baseline repeatability	±5 µW *Electric cooling specifications: -50°C to 300°C, 10°C/min	
Programmable Rate	0.01°C to 100°C/min	
Purge gas control	Built-in Mass flow controller (2 gas lines)	
Auto Sampler *Optional	50 positions	
Electrical Cooling Unit *Optional	-80°C to 500°C	
Auto LN_2 Gas Cooling Unit *Optional	-150°C to 725°C	
Auto Air Cooling Unit *Optional	Ambient to 725°C	

Dimensional information *Excluding protrusions



Science for a better tomorrow

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⚠ Notes regarding safety

Be sure to read the operation manual before use in order to use the product safely.

- Prices vary depending on the specifications and configuration.
- The external appearance and some of the specifications of the products described in this catalog are subject to change without prior notice.
- "Real View" is a registered trademark of Hitachi High-Tech Science Corporation in the US, the EU, the UK, China, India, Turkey, Brazil and Japan.
- "NEXTA" is a registered trademark of Hitachi High-Tech Science Corporation in the US, the EU, the UK, Korea, Taiwan and Japan.
- The PC screenshots in this catalog are inserted images.

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Differential Scanning Calorimeter (DSC)



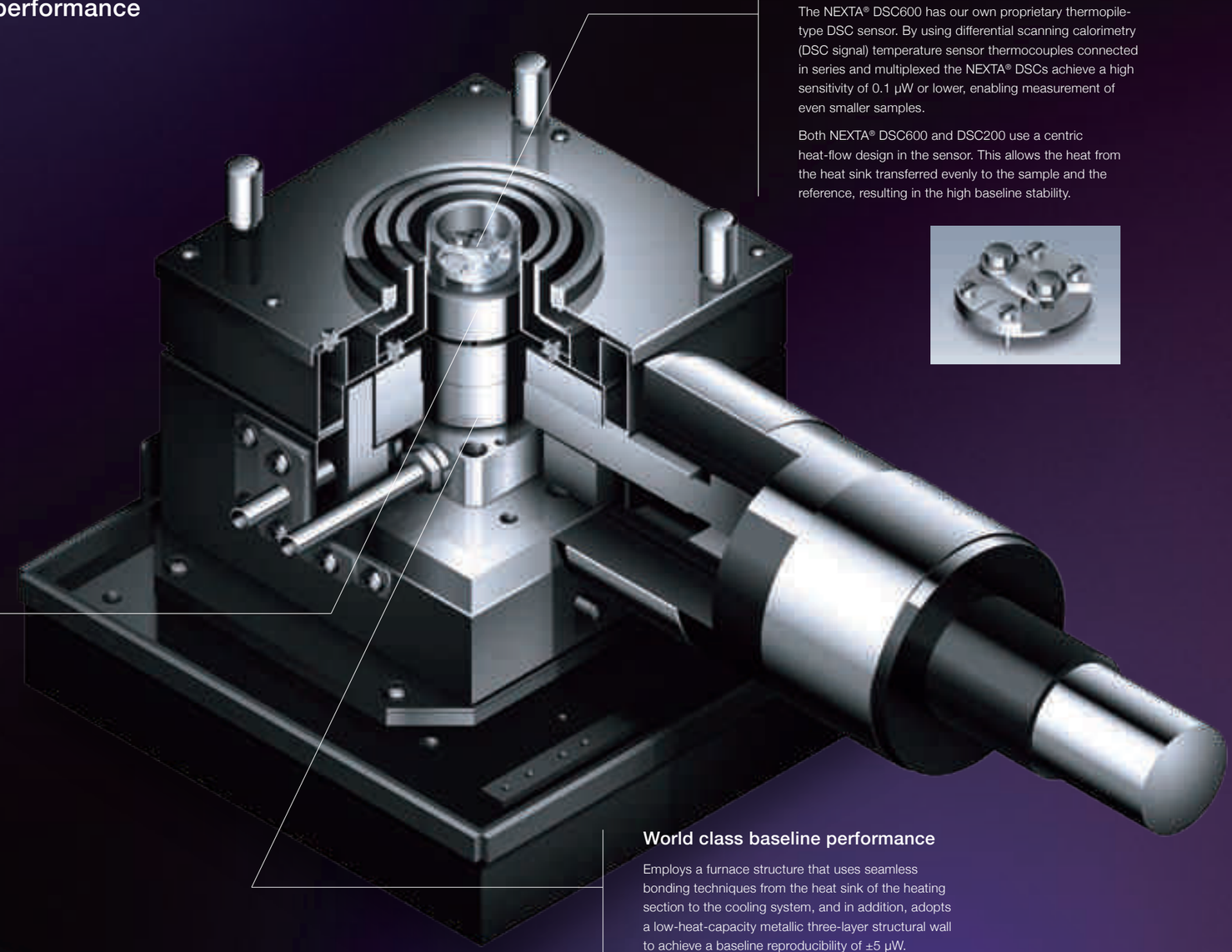
Science for
a better tomorrow

Hitachi High-Tech Science

To the NEXT stage in thermal analysis

Leading-edge materials evaluation supported by leading-edge technology.
NEXTA® DSC achieves world class sensitivity and baseline performance

Hitachi High-Tech Science, responding to a variety of thermophysical property assessment needs from thermal analysis to viscoelasticity, has released the new NEXTA® series of instruments.
The NEXTA® DSC (Differential Scanning Calorimeter) achieves extremely stable baseline reproducibility through a sensor configuration that employs a central heat flux system together with a proprietary sensor to attain high sensitivity by a seamless furnace structure, metallic three-layer insulation structure, and other innovations. The series of DSCs provide cutting-edge measurement technology for quality control in various fields as well as research and development of polymers, inorganic materials, pharmaceuticals, and others.



Novel sensor design that achieves high sensitivity

The NEXTA® DSC600 has our own proprietary thermopile-type DSC sensor. By using differential scanning calorimetry (DSC signal) temperature sensor thermocouples connected in series and multiplexed the NEXTA® DSCs achieve a high sensitivity of 0.1 μW or lower, enabling measurement of even smaller samples.

Both NEXTA® DSC600 and DSC200 use a centric heat-flow design in the sensor. This allows the heat from the heat sink transferred evenly to the sample and the reference, resulting in the high baseline stability.



High temperature accuracy and precision

Increases thermal uniformity to the utmost limit by making the heat sink from a high-thermal-conductivity, high-purity Ag block. In addition, high temperature and calorimetric accuracy are achieved with micro-analog signal processing technology.

World class baseline performance

Employs a furnace structure that uses seamless bonding techniques from the heat sink of the heating section to the cooling system, and in addition, adopts a low-heat-capacity metallic three-layer structural wall to achieve a baseline reproducibility of ±5 μW.

Differential Scanning Calorimeter(DSC)

NEXTA® DSC Series

DSC600 / DSC200

A lineup of high-sensitvity/high-resolution top- end model and high-sensitvity standard model

Types of DSC sensor

The 600 Sensor and the 200 Sensor are installed in DSC600 and DSC200, respectively. With these advanced sensors, NEXTA® DSC series achive high sensitivity together with other superior chracteristics.

	600 Sensor	200 Sensor
Model	DSC600	DSC200
Type	High sensitivity/High resolution	High sensitivity
Thermocouple	Multi	Single
Sensor holder material	Constantan	Au



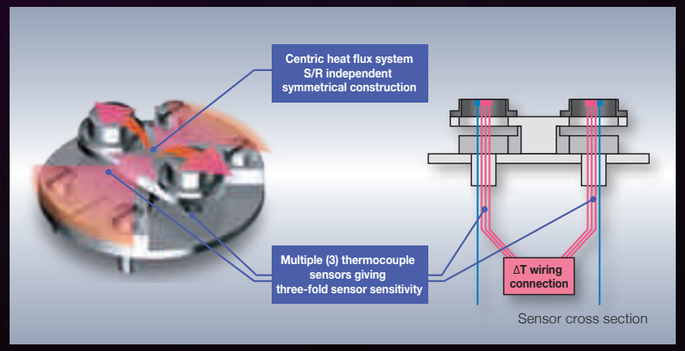
600 Sensor



200 Sensor

DSC600

DSC600 is equipped with a proprietary thermopile sensor containing multiple thermocouples, allowing it to achieve high resolution and world class sensitivity. In addition to reliably detecting minute phase transitions, the DSC600 is a high-end model that can handle a variety of measurement needs.



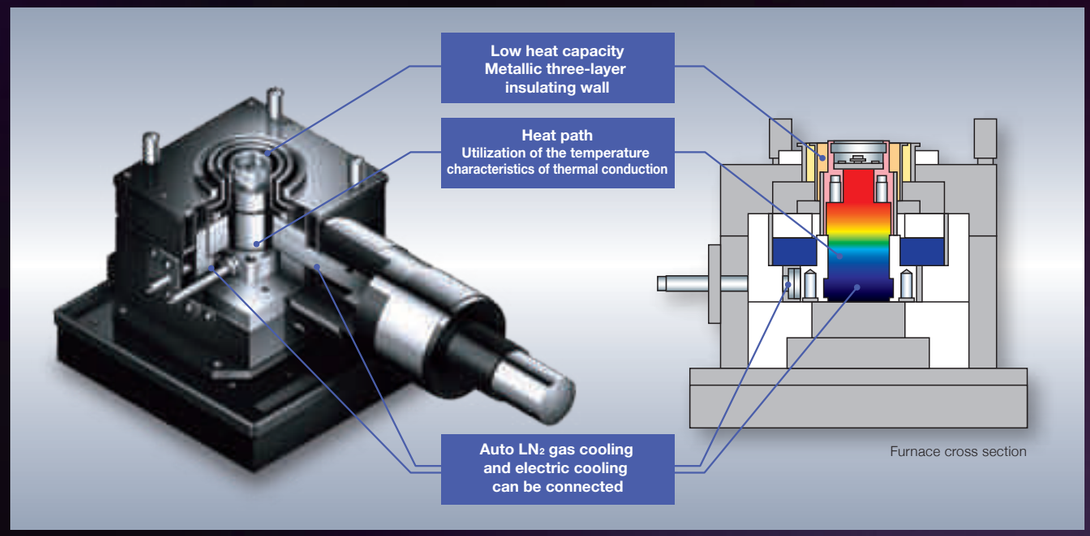
Sensor cross section

DSC200

DSC200 uses a newly developed Au sensor which delivers high calorimetric precision with high cost performance.
The centric heat flux method used also in DSC600 provide high baseline stability.

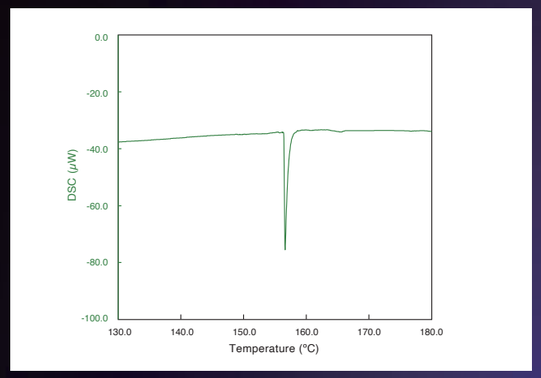
Three-layer insulation-type furnace

The heating furnace achieves world class baseline reproducibility and high stability through a low-heat-capacity metallic three-layer insulation structure.
In addition, superior Temperature control is achieved in each cooling system by utilizing the temperature characteristics of the heat-path materials from the heat sink to the cooling block.

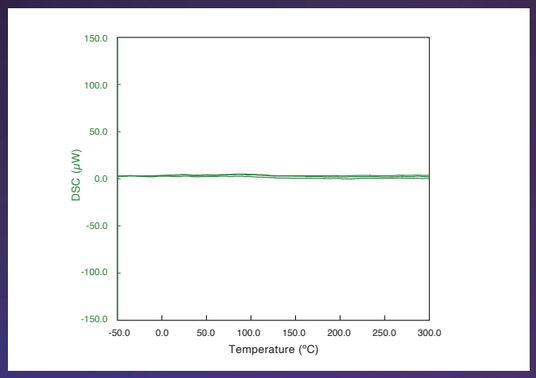


Furnace cross section

Sensitivity and baseline reproducibility data



Indium 5 μg



Repeated measurements of baseline(n=3)

Cooling Unit / Automation / Expandability

Realizes highly efficient measurements for a wide variety of needs from material development to product evaluation

Selectable cooling unit (option)

Three cooling options are provided. These can be selected according to the usage environment and application.
With the NEXTA® DSC, it is possible to simultaneously connect Electrical cooling unit and Auto LN_2 gas cooling unit. This allows one to use electrical cooling under normal conitions, which can hold down running costs, and to use LN_2 gas cooling only when cooling down to further lower temperatures.

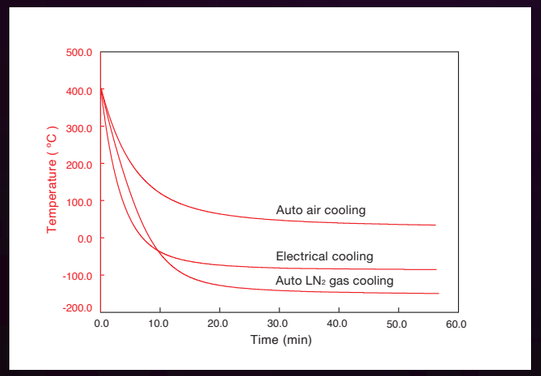


Electrical cooling unit

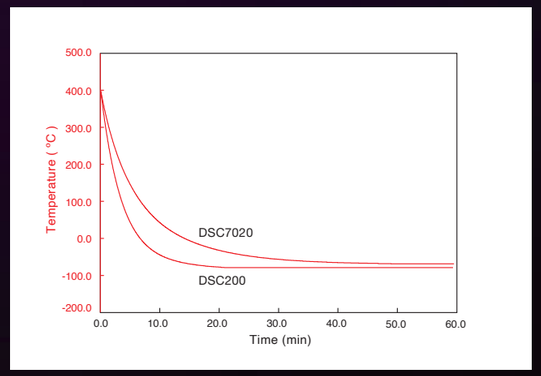


Auto LN_2 gas cooling unit

Cooling system	Electrical cooling unit	Auto LN_2 gas cooling unit	Auto air cooling unit
Temperature range	-80°C to 500°C	-150°C to 725°C	Room temperature to 725°C
Required refrigerant	No refilling required	Liquid nitrogen	Compressed air
Other	100-115V 220-230V	Liquid nitrogen tank30L / 50L / 100L	Used for cooling after measurements are completed (built into main unit)



Cooling profile



Comparison of electrical cooling performance

Auto Lid unit (option)

A new optional Auto Lid unit is available.
This automatically opens and closes the sample unit cover and also locks when measurements are in progress and when the interior is at a high temperature. This makes using the instrument safer as well as improving performance.



Auto sampler / Safety cover (option)

Delivering automatic analysis with highly efficient measurement through a reliable and smooth transport

The Hitachi High-Tech Science autosampler has an established repuation for providing reliable and stable sample transport performance, and this technology has been adapted to the NEXTA® series. It offers secure transport by gripping the sample pan firmly with 4 fingers, and it delivers high throughput when used in combination with the auto-analysis software function.
An autosampler safety cover is provided as an option for increasing the reliablity and safety of measurements.



Sample pan transported by Auto sampler(up to 50 samples)

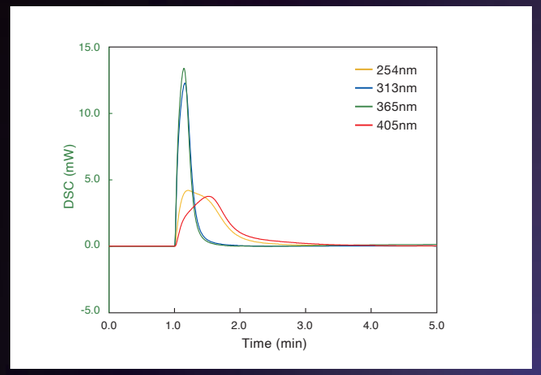


Auto sampler safety cover

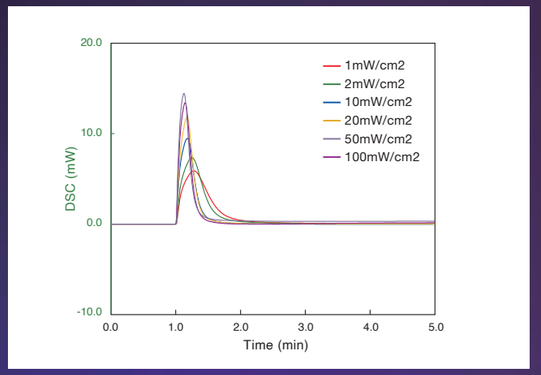
PDC (option)

A photocalorimeter measures a sample while irradiating it with light (PDC) and is an option used for evaluating the curing reaction of a photosetting resin or resist. The curing rate of a photosetting resin used on a production line can be evaluated by varying the wavelength and intensity of the light irradiation. This option can be used in the material developent and quality control as well as for evaluation of production processes that use such materials.





Reaction dependence on irradiation wavelength



Dependence on irradiation intensity

High Functionality / Multiple Applications

Visualization of sample changes with Real View® and separate measure ment of overlapping phase transitions through temperature modulated DSC.

Real View® sample observation and thermal analysis

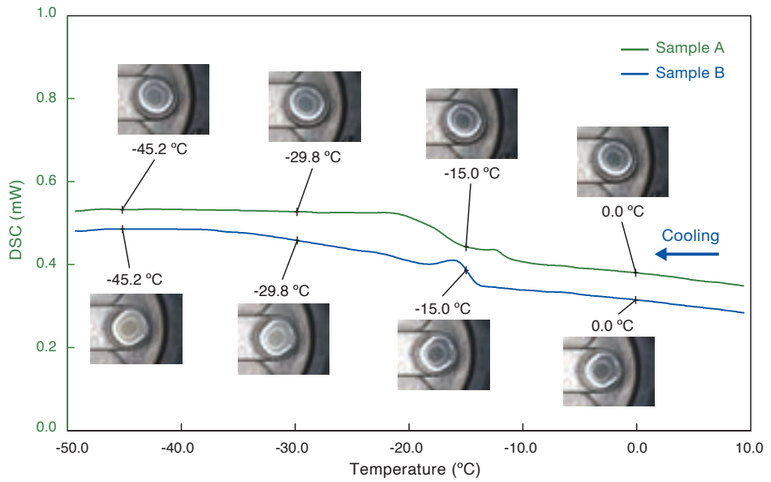
"Visibility" for better "Measurement"

With conventional thermal analysis, it has not been possible to observe a sample while a measurement is in progress. With Real View®, however, it is possible to observe changes in the sample condition while a measurement is in progress, using continuous images obtained with a high-resolution camera. More minute changes can be captured by employing the digital zoom function. Since thermal analysis data are indexed to sample image data, another level of data analysis becomes possible. Moreover, image analyses can be carried out using thermal analysis software, including a digital zoom, and it is possible to conduct a greater variety of analyses such as length measurements and color analysis.



Color changes during cooling for two types of engine oil

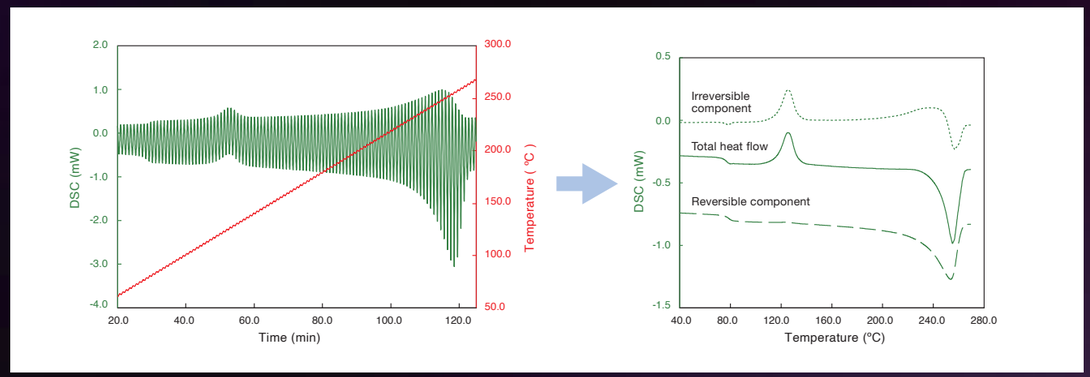
With the NEXTA® DSC series, it is possible to make Real View® measurements down to -50°C.
In cloud point measurements of two types of engine oil, the crystallization(exothermic peak) occurs at higher temperature on sample A.
It can be seen that while sample B becomes cloudier from -30°C, the color change for sample A is small.



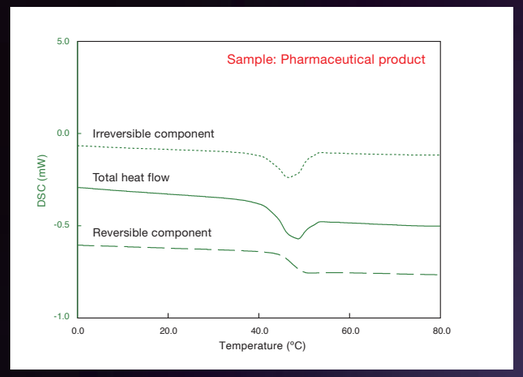
Temperature modulated DSC

Temperature modulated DSC is a measurement technique that separates overlapping phase transitions by adding a sinusoidal variation to a constant rate of temperature increase or decrease. This allows three components are calculated from the results of temperature modulated DSC measurements.

- **Total heat flow**
- **Specific heat component (reversible component)**
- **Irreversible component**

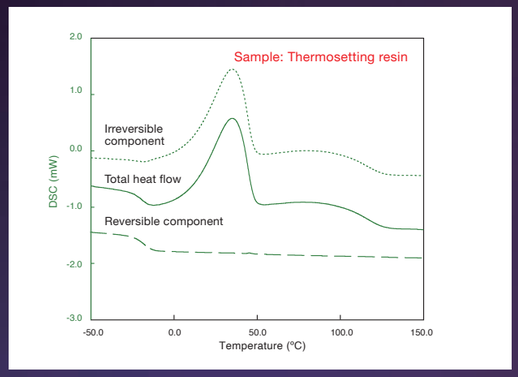


Compared to standard DSC measurements, in cases where the reversible component (glass transition) and irreversible components (enthalpy relaxation, curing, crystallization, evaporation, decomposition, etc.) overlap each other, these components can be separated by temperature modulated DSC.



Separation of glass transition and enthalpy relaxation

This is effective for evaluation of the glass transition accompanied by enthalpy relaxation of polymeric materials, the stability of pharmaceutical products in the presence of storage temperature over time, and other such applications.



Separation of glass transition and curing reaction

Overlapping glass transitions and curing reactions can be separated by this technique. Thermosetting resins are widely utilized in electronic components, aircraft, automobiles, and the like.

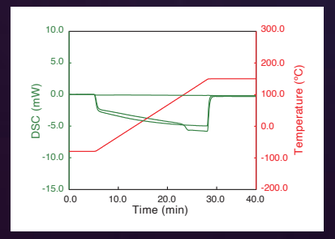
Optional Software / Sample Pans

We provide software for specific heat, kinetics, and CFR21 Part 11 compliance as well as an extensive lineup of sample pans and electrical and manual sample sealers.

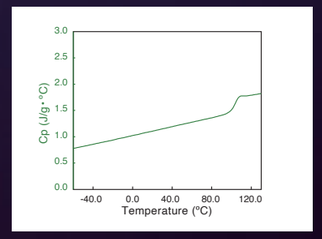
Advanced software

Specific heat calculation

This software calculates the specific heat capacity from baseline, standard reference material, and sample data.



Three data items



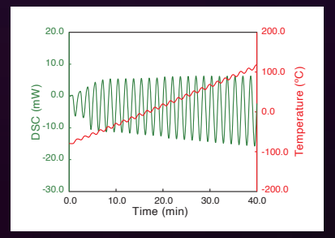
Cp calculation

Temperature modulated DSC

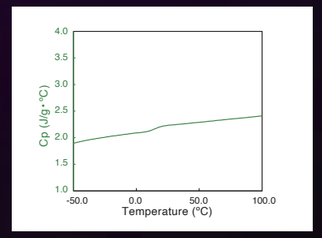
This software transforms temperature modulated DSC results into reversible and irreversible components.

CP determination by temperature modulated DSC method

With the NEXTA® DSC series, it is also possible to make specific heat capacity measurements by temperature modulation.



Temperature modulation



Cp calculation

Kinetic analysis

This software calculates the activation energy and thermal degradation time (lifetime) from data of different heating rates. This is used in polymeric material basic research and durability estimates.

Purity analysis

This software calculates the purity of a sample by evaluating its melting peak.

CFR21 Part 11

This software handles access control to systems and data, electronic signatures for data, audit trails, and other requirements of CFR21 Part 11.

Waveform separation simulation

This software separates peaks by simulation when two peaks overlap.

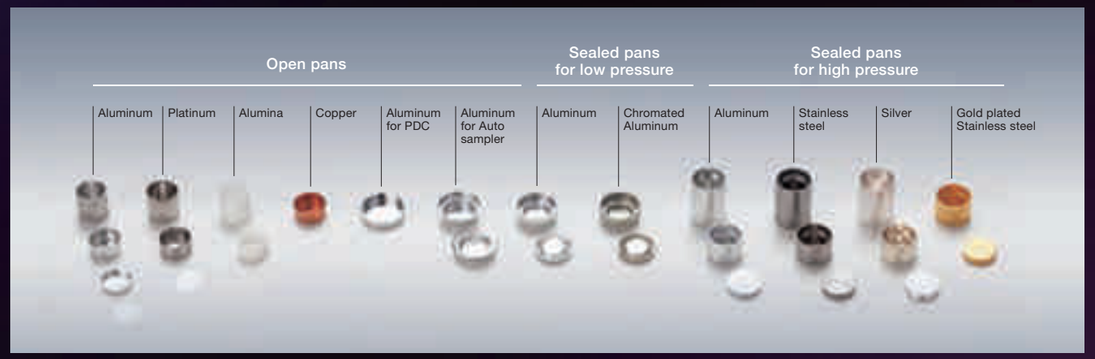
Automated analysis

This software automates data analysis, output, and storage of results, allowing the operation time required for analysis can be shortened.

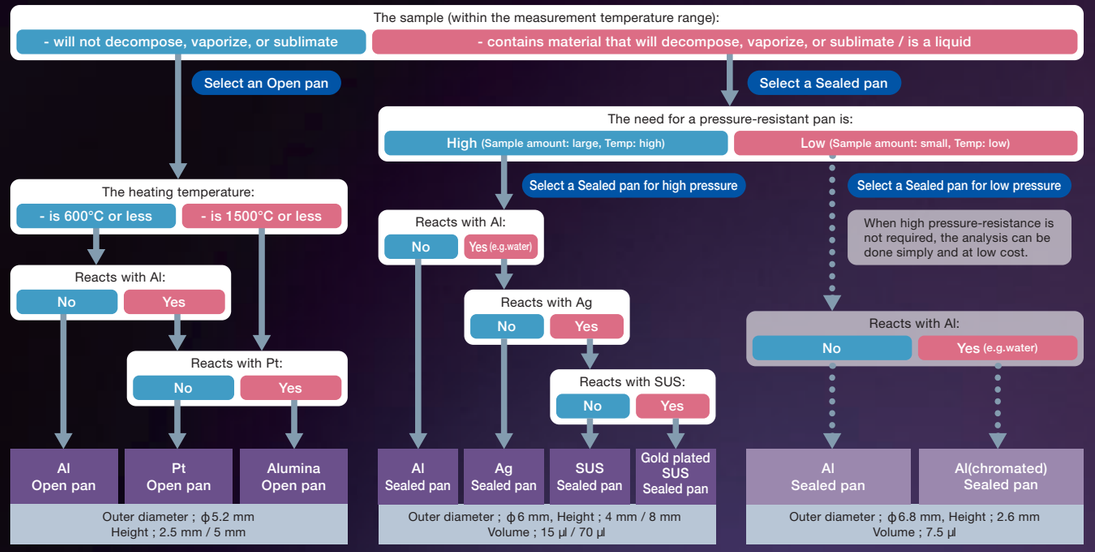
External data import

External text data can be imported for data analysis on the analysis software.

Sample pans



Sample pan selection flowchart



A sample sealer is used when crimping or sealing a sample pan. A manually operated sample sealer is standard; however, with sealed pans, here are risks depending on the sealing pressure that data may change, the sample may boil over when the measurement is in progress, and other such risks.
An electrical sample sealer is a convenient sealer that is used for the sealed pans above so as to always seal at a constant pressure.



Electrical sample sealer



Manual sample sealer

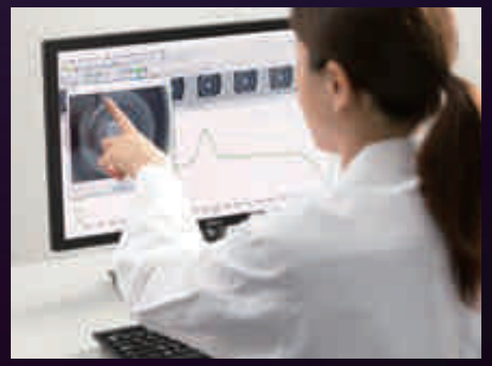
Soft ware

The software for NEXTA® can be easily operated to supp ort everything from simple use to complex applications.

The software offers a variety of usage methods for different user levels, including a guidance function that enables beginners to perform measurements according to standard test methods such as JIS, ISO, ASTM etc. A simple measurement mode that is optimized for routine measurements and operation via tablet is also available.

- **Guidance function that provides step-by-step instructions from measurement to analysis**
- **Simple measurement mode that delivers an operating environment suitable for routine measurements**
- **Touch panel operation and tablet support**





Guidance Function

-Standard-Based Guidance

With Standard-Based Guidance, users will be walked-through the processes to perform standard methods measurements according to organizations such as JIS, ISO, ASTM etc. Conditions such as sample weight, temperature range, scanning rate, gas flow rate, etc. called out in standards are set as default measurement suggestions.□

-Calibration Guidance

Users will be walked through the process to calibrate the instrument. For temperature calibration, analysis of data will be done automatically, and analyzed data will be used to adjust calibration automatically. Users can calibrate easily and correctly.

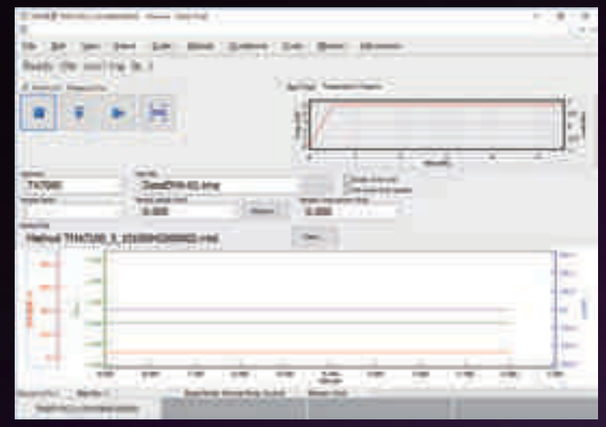
-Custom-Made Guidance

Users can create their own guided standard methods, so anyone can measure samples easily (and correctly). By following explanations in the custom-made guidance, such as sample preparation method, sample weight, selection of sample container, etc., different operators can make the same types of measurements repeatedly.



Simple mode

In simple mode, by setting measurement conditions in advance, users just need to input minimum sample information in one screen and the measurement can be started instantly. Using this mode not only helps minimize complicated operations, such as setting up and checking detailed experimental conditions, but it also reduces both total process time and operator entry mistakes.



Touching Operation

The software for NEXTA® supports touchscreen operation on PC and tablets. By using portable devices, even if users are at a remote location, they can still check the progress of the measurement and analyze the acquired data.



Other Functions and Features

Many other user-friendly functions and features are included which meet a wide variety of user needs. The reporting feature enables the delivery of multiple data types at the same time as well as generating customizing report formats.

- **Multi-customization function**
- **Active scale**
- **Multiple output function**
- **Free license**

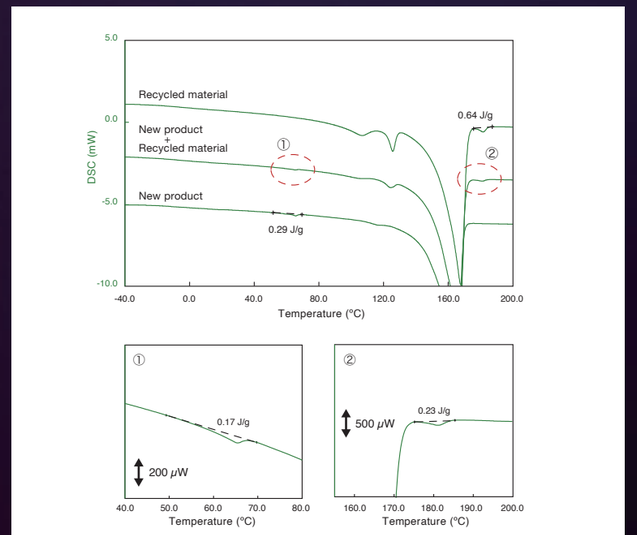


Application Data Collection

Available application notes including polymeric materials, minute sample quantities, pharmaceuticals, food products, and other applications.

Recycled polymer (PP)

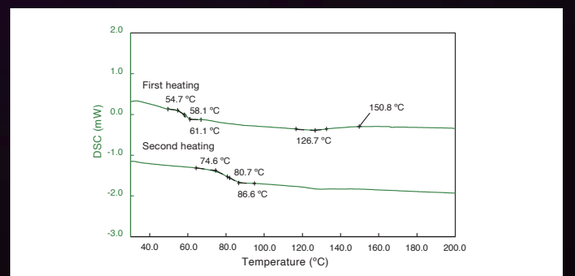
Multiple DSC peaks can be seen other than PP melting near 160°C in PP recycled material that was produced by recycling. A very small endothermic peak can be seen from 60 to 70°C even in newly produced PP that is not recycled. In PP that contains recycled material, peaks originating from both the recycled material and new product can be seen, and the mixing ratio of recycled material can be estimated from the peak intensities.

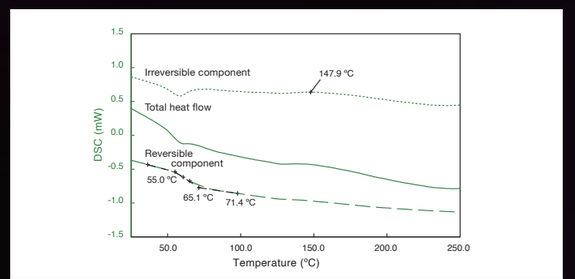


CFRP Normal measurement and TMDSC measurement

Because the amount of epoxy resin in CFRP is small, excellent sensitivity and baseline stability are required to measure the glass transition and exothermic heat associated with the uncured portion.
The glass transition was seen in the neighborhood of 50 to 60°C during the first heating and in the neighborhood of 70 to 90°C during the second heating. During the first heating, a change was seen from 126.7°C in the exothermic heat direction.

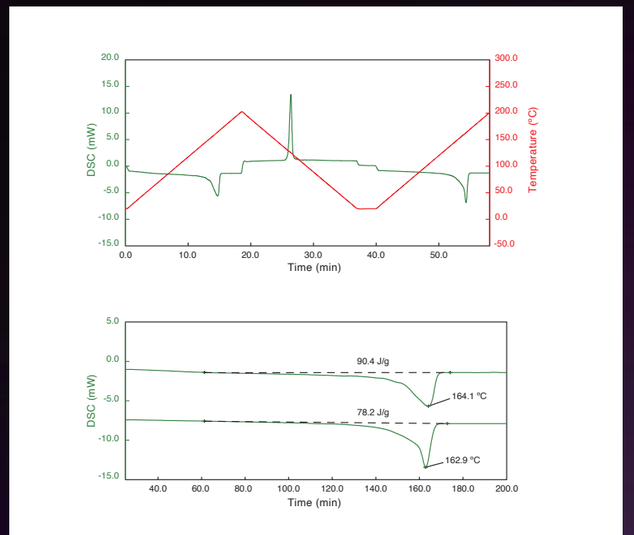
These results were obtained by temperature modulated DSC. The endothermic peak due to the glass transition became clearly visible in the separated reversible component, and the exothermic heat due to curing was separated in the irreversible component. It can be seen that the glass transition temperature during the second heating increased due to curing.



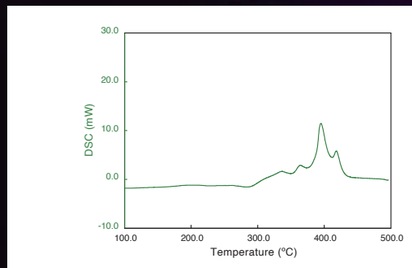


CNF+PP 1st and 2nd thermal history

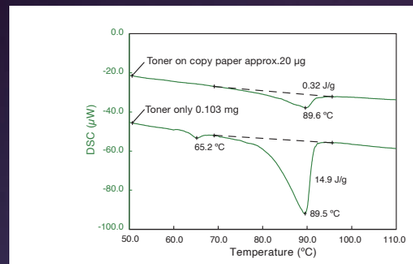
The measurement results for the transition temperatures of plastic are affected by the thermal history. For that reason, the measurement results during the second heating after cooling can be different than those during the first heating. When it is desired to compare samples having different temperature histories under the same conditions, cooling after the first heating is carried out at a constant controlled cooling rate, and the physical properties are evaluated during the second heating.
The method for this sort of constant-rate heating / cooling / heating measurement is described in ASTM, ISO, and other standards.



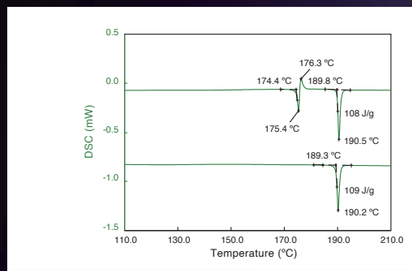
Battery material / NMC + Electrolyte solvent



Toner



Pharmaceutical product / Carbamazepine



Protein / Aqueous lysozyme solution

